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INFORMATION PROCESSING DEVICE, INFORMATION PROCESSING SYSTEM, INFORMATION PROCESSING METHOD, AND COMPUTER PROGRAM PRODUCT

Abstract

According to an embodiment, an information processing device includes one or more hardware processors. The one or more hardware processors are configured to: acquire a first stream transmitted from a transmitting device; and relay the acquired first stream to both a first server device and a second server device when a sending request requesting to send a stream from the second server device to a receiving device is received, and relay the acquired first stream to the first server device when the sending request is not received, the first server device being configured to store an input stream into a storage device, the second server device being configured to send a stream with a delay time not greater than a second delay time that is less than a first delay time with which a stream stored in the storage device is distributed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-025139, filed on Feb. 22, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to an information processing device, an information processing system, an information processing method, and a computer program product.

BACKGROUND

[0003] Systems for collecting and storing videos (streams) captured using a camera (image capturing device), and distributing the videos to receiving devices have been known. As an example of such a system, a technique for switching a mode between a high-quality mode for performing high-quality encoding of the stream, and a low-latency mode for sending the stream with a less delay time has been proposed. The low-latency mode is used when camera operations such as turning, zooming, and focusing are performed.

[0004] With the technique described above, while the low-latency mode is in use, the high-quality stream distribution stops. In other words, one of a stream in the high-quality mode and a stream in the low-latency mode can be sent. Streams in both of these modes can be sent if a system for the high-quality mode and a system for the low-latency mode are provided, but issues such as an increase in data transmission quantity may arise, as a result of causing the camera to send the stream to the two systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a block diagram of an information processing system according to a first embodiment;

[0006] FIG. 2 is a flowchart of a collecting process in the first embodiment;

[0007] FIG. 3 is a flowchart of a large-scale streaming process during a normal mode in the first embodiment;

[0008] FIG. 4 is a flowchart of a parallel mode switching process in the first embodiment;

[0009] FIG. 5 is a flowchart of a process for returning to the normal mode in the first embodiment;

[0010] FIG. 6 is a block diagram of an information processing system according to a second embodiment;

[0011] FIG. 7 is a flowchart of a parallel mode switching process in the second embodiment;

[0012] FIG. 8 is a flowchart of a process for returning to the normal mode in the second embodiment;

[0013] FIG. 9 is a block diagram of an information processing system according to a first modification; and

[0014] FIG. 10 is a hardware configuration diagram of a device according to the first or second

embodiment.

DETAILED DESCRIPTION

[0015] According to an embodiment, an information processing device includes one or more hardware processors. The one or more hardware processors are configured to: acquire a first stream transmitted from a transmitting device; and relay the acquired first stream to both a first server device and a second server device when a sending request requesting to send a stream from the second server device to a receiving device is received, and relay the acquired first stream to the first server device when the sending request is not received, the first server device being configured to store an input stream into a storage device, the second server device being configured to send a stream with a delay time not greater than a second delay time that is less than a first delay time with which a stream stored in the storage device is distributed.

[0016] Exemplary embodiments of an information processing device according to the present invention will now be explained in detail with reference to the accompanying drawings.

First Embodiment

[0017] An information processing system according to a first embodiment includes: a video server for distributing a stream; a low-latency server for sending a stream with a less delay time; and an information processing device configured to relay a stream received from a camera to the video server and the low-latency server. The information processing device relays the stream received from the camera to both of the video server and the low-latency server only while there is an access to the low-latency server. When there is no access to the low-latency server, the information processing device relays the stream received from the camera only to the video server. In this manner, it becomes possible to continue the distribution of a stream while suppressing an increase in the data transmission quantity from the camera, for example.

[0018] For example, there are times when it is necessary to receive a stream with a less delay time for the purpose such as to adjust the angular field of a camera, and the like. In such a case, with the technique for switching the high-quality mode to the low-latency mode, for example, the distribution of the high-quality stream stops. In this embodiment, however, the distribution of the high-quality stream can be performed, in parallel with the reception of the low-latency stream.

[0019] FIG. 1 is a block diagram illustrating one example of a configuration of the information processing system according to the first embodiment. As illustrated in FIG. 1, the information processing system according this embodiment includes an information processing device **100**, a cloud **200**, a plurality of receiving devices **300a**, **300b**, and **300c**, a receiving device **400**, and a camera **500**.

[0020] Because the receiving devices **300a**, **300b**, **300c** have the same configuration, when it is not necessary to distinguish one from the others, the receiving devices **300a**, **300b**, and **300c** will be simply referred to as receiving devices **300**. The number of the receiving devices **300** is not limited to three, and may be two, or four or more. The number of the other devices are also not limited to those illustrated in FIG. 1. For example, the information processing system may include a plurality of cameras **500**, or may include a plurality of receiving devices **400**.

[0021] The devices included in the information processing system (the information processing device **100**, the cloud **200**, the receiving devices **300**, the receiving device **400**, and the camera **500**) are connected to one another over a network such as the Internet, which is not illustrated in FIG. 1. The network may be any one of a wired network, a wireless network, and a hybrid network including wired and wireless networks.

[0022] The camera **500** corresponds to an image capturing device configured to capture a video. The camera **500** also corresponds to a transmitting device configured to transmit a video captured thereby to the information processing device **100**. The transmitting device is not limited to the camera **500** (image capturing device). The camera **500** includes an encoding unit **501** and a transmitting unit **502**.

[0023] The encoding unit **501** encodes a captured video. The encoding unit **501** may use any

encoding scheme to encode the video. The transmitting unit **502** transmits the encoded video (stream) to the information processing device **100**.

[0024] The cloud **200** includes a video server **210** (first server device), a low-latency server **220** (second server device), and a web server **230** (third server device). It means that the cloud **200** implements these three servers on the cloud environment. Some of the three servers may be built on a cloud environment that is different from the environment where the others are built. The method in which the three servers are built is not limited to the method for building the servers on the cloud environment. For example, the three servers may, partly or entirely, be built as independent server devices that are not in the cloud environment.

[0025] The video server **210** includes a storage device **211**, and stores a stream input thereto in the storage device **211**. For example, the video server **210** receives a stream from the information processing device **100**, and stores the received stream in the storage device **211** as a file.

[0026] The file may have any data structure, and some examples of the structure are described below. A duration put into a file and data format may be combined in any way. [0027] Duration put into a file: a structure in which every frame is put into a file, and a structure in which every segment is put into a file. [0028] Data format: a structure in which a network abstraction layer (NAL) stream is retained as it is, or a structure in which the stream is converted into a format such as MP4.

[0029] The storage device **211** may be implemented as any storage medium generally used, such as a flash memory, a memory card, a random access memory (RAM), a hard disk drive (HDD), and an optical disk.

[0030] The low-latency server **220** is a server device configured to send a stream with a delay time smaller than that of the video server **210**. For example, the low-latency server **220** sends the stream received from the information processing device **100** with a delay time not greater than a delay time **T2** (second delay time) that is less than a delay time **T1** (first delay time) on the distribution of the stream stored in the storage device **211** of the video server **210**.

[0031] The web server **230** is a server that distributes the stream stored in the storage device **211** in the video server **210**. For example, the web server **230** acquires the stream from the storage device **211** on the video server **210**, and distributes the acquired stream to the receiving devices **300**.

[0032] Each of the receiving devices **300** is a device configured to receive the stream distributed from the web server **230**, to decode the received stream, and to display the decoded stream. For example, when the receiving device **300** accesses an address (e.g., uniform resource locator (URL)) for streaming, the receiving device **300** receives a viewer for displaying the stream from the web server **230**. In the explanation hereunder, the address from which the viewer for streaming is acquired will be sometimes referred to as a large-scale streaming URL. This viewer has a function for receiving and displaying the stream from the web server **230**. The receiving device **300** decodes and displays the distributed stream, using the received viewer. The viewer may also be set to the receiving device **300** in advance, instead of being sent from the web server **230**.

[0033] For the distribution of a stream between the web server **230** and the viewer, HTTP Live Streaming (HLS), or Dynamic Adaptive Streaming over HTTP (DASH) can be used, for example. HLS and DASH may become high in latency because of the following functions: [0034] A stream is distributed as a file referred to as a segment, which is an accumulation of the stream over a certain length of time. [0035] A stream is replayed after the viewer buffers two segments so that there is no interruption in the stream being replayed. [0036] The web server **230** sends the stream upon being requested by the viewer.

[0037] The receiving device **400** is a device configured to receive a stream sent from the low-latency server **220**, to decode the received stream, and to display the stream. For example, when the receiving device **400** accesses an address (for example, URL) for a low-latency stream sending, the receiving device **400** receives a viewer for displaying a stream (hereinafter, a low-latency viewer) from the web server **230**. The low-latency viewer may also be received from the low-latency server

220. The low-latency viewer may be set to the receiving device **400** in advance, instead of being sent from the web server **230**. In the explanation hereunder, the address for acquiring the low-latency viewer will be sometimes referred to a low-latency URL. The low-latency viewer has a function for receiving a stream from the low-latency server **220** and to display the stream, with a less delay time. The receiving device **400** decodes and displays the distributed stream, using the received low-latency viewer.

[0038] For the stream sending between the low-latency server **220** and the low-latency viewer, for example, Web Real-Time Communication (WebRTC) can be used. Although WebRTC is not suitable for large-scale streaming, WebRTC can reduce latency, because of the functions to be described below. The reason why WebRTC is not suitable for large-scale streaming is that the distribution load increases as the number of distribution targets increases. [0039] It is not necessary to wait for the stream corresponding to one segment before putting the stream into a file. [0040] It is not necessary for the viewer to wait for the stream corresponding to two segments to become buffered. [0041] It is not necessary to wait for a request from the viewer before sending the stream. [0042] Generally, a plurality of separate systems are used depending on the applications, e.g., a system using HLS and DASH for the large-scale streaming, and a system using WebRTC for the low-latency transmit. However, with such a configuration, it is necessary for the camera to transmit a stream to the plurality of systems, and therefore, the data transmission quantity may increase. [0043] By contrast, in this embodiment, the information processing device **100** relays the same stream sent from the camera to both of the web server **230** for distributing the stream over HLS or DASH, and the low-latency server **220** for transmitting the stream over WebRTC. In this manner, both of the large-scale streaming and the low-latency transmit can be supported.

[0044] The information processing device **100** includes a processing unit **110**. The processing unit **110** includes an acquiring unit **111**, a relaying unit **112**, and a switching unit **113**. Physically, the information processing device **100** may be implemented as a single device or a plurality of devices. For example, the information processing device **100** may be built on a cloud environment that is the same as the cloud **200** or different from the cloud **200**.

[0045] The acquiring unit **111** acquires various types of data from external devices such as the camera **500**. For example, the acquiring unit **111** acquires a stream ST1 (first stream) transmitted from the camera **500**.

[0046] The relaying unit **112** relays the acquired stream ST1 to the video server **210** and the low-latency server **220**. For example, when the switching unit **113** receives a sending request requesting to send a stream from the low-latency server **220** to the receiving device **400**, the relaying unit **112** relays the stream ST1 to both of the video server **210** and the low-latency server **220**. In the explanation hereunder, the mode for relaying the stream ST1 to both of the video server **210** and the low-latency server **220** will be sometimes referred to as a parallel mode.

[0047] When no sending request requesting to send a stream from the low-latency server **220** to the receiving device **400** has been received, the relaying unit **112** relays the stream ST1 to the video server **210** but not to the low-latency server **220**. In the explanation hereunder, the mode in which the stream ST1 is relayed to the video server **210** but not to the low-latency server **220** will be sometimes referred to as a normal mode.

[0048] A sending request is sent from the receiving device **400** to the information processing device **100** via the low-latency server **220**, for example. The receiving device **400** sends a stream sending request to the low-latency server **220** using the low-latency viewer, for example. The low-latency server **220** sends the sending request sent from the low-latency viewer to the information processing device **100**.

[0049] The relaying unit **112** can be configured to operate in the normal mode or the parallel mode, in accordance with the mode to which the switching unit **113** is switched.

[0050] In other words, the switching unit **113** switches the relaying mode for the relaying unit **112** in response to a sending request received from the low-latency server **220**. For example, when a

sending request is received from the low-latency server **220**, the switching unit **113** switches the mode of the relaying operation of the relaying unit **112**, from the normal mode to the parallel mode. When the access from the low-latency viewer is ended, the switching unit **113** switches the mode of the relaying operation of the relaying unit **112**, from the parallel mode to the normal mode. In the manner described above, the parallel mode can be said to be a temporary mode transitioned only when there is an access from the low-latency viewer.

[0051] The processing unit **110** is implemented with one or more hardware processors, for example. The processing unit **110** may be implemented by causing a processor such as a central processing unit (CPU) or a graphics processing unit (GPU) to execute a computer program, that is, may be implemented with software. The processing unit **110** may be implemented as a processor such as a dedicated integrated circuit (IC), that is, the processing unit **110** may be implemented with hardware. The processing unit **110** may also be implemented using software and hardware in combination. When a plurality of hardware processors are used, one, or two or more of the units included in the processing unit **110** (the acquiring unit **111**, the relaying unit **112**, the switching unit **113**) may be implemented by the hardware processors.

[0052] Various processes performed by the information processing system according to the first embodiment will now be explained with reference to FIGS. **2** to **5**. FIG. **2** is a flowchart giving an outline of a collecting process in the first embodiment. The collecting process is a process for collecting a video captured by the camera **500** onto the video server **210**, and storing the video in the video server **210**. FIG. **2** corresponds to an example in which the relaying unit **112** operates in the normal mode. For example, the relaying unit **112** is set to the normal mode during an initial state after the information processing device **100** is started.

[0053] The encoding unit **501** in the camera **500** encodes a video captured by the camera **500** (Step **S101**). The transmitting unit **502** then transmits the encoded video (stream) to the information processing device **100** (Step **S102**).

[0054] The acquiring unit **111** in the information processing device **100** acquires the stream having been transmitted from the camera **500**, and passes the stream to the relaying unit **112**. The relaying unit **112** relays the received stream only to the video server **210** (Step **S103**).

[0055] The video server **210** stores the received stream in the storage device **211** as a file (Step **S104**). For example, the video server **210** waits until the stream corresponding to the duration of a stream to be put into a file is received (e.g., one frame or one segment), converts the stream into a file by performing a format conversion as required, and stores the resultant file in the storage device **211**. When the camera **500** is configured to send a stream in units of a segment, the video server **210** does not need to wait until the stream corresponding to the duration of a stream to be put into a file is received.

[0056] An outline of the large-scale streaming process in the normal mode will now be explained with reference to FIG. **3**. FIG. **3** is a flowchart giving the outline of the large-scale streaming process during a normal mode in the first embodiment.

[0057] A user of the receiving device **300**, for example, designates the large-scale streaming URL using a browser or the like. The receiving device **300** accesses the web server **230** in accordance with the designation of the large-scale streaming URL (Step **S201**).

[0058] The web server **230** sends a viewer to the receiving device **300** as an access source (Step **S202**).

[0059] Using the sent viewer, the receiving device **300** sends a request for a stream to the web server **230** regularly, for example (Step **S203**).

[0060] The web server **230** distributes the stream requested by the viewer to the receiving device **300** that is a request source (Step **S204**). Although not illustrated in FIG. **3**, for example, the web server **230** reads the requested stream from the storage device **211** of the video server **210**, and distributes the read stream to the receiving device **300**.

[0061] The receiving devices **300** receives the stream from the web server **230** and decodes the

stream using the viewer, and displays the decoded stream on a display device provided to the receiving device **300**, for example (Step **S205**).

[0062] An outline of a process for switching to the parallel mode (hereinafter, a parallel mode switching process) will now be explained with reference to FIG. **4**. It is different from FIG. **2** in that the stream is relayed not only to the video server **210** but also to the low-latency server **220** if the mode is switched to the parallel mode. FIG. **4** is a flowchart giving the outline of the parallel mode switching process in the first embodiment.

[0063] For example, a user of the receiving device **400** designates the low-latency URL using a browser or the like. The receiving device **400** accesses the web server **230** in accordance with the designation of the low-latency URL (Step **S301**).

[0064] The web server **230** sends a low-latency viewer to the receiving device **400** that is an access source (Step **S302**).

[0065] The receiving device **400** accesses the low-latency server **220** using the low-latency viewer sent thereto, and establishes a connection with the low-latency server **220** (Step **S303**).

[0066] The low-latency server **220** notifies the switching unit **113** in the information processing device **100**, of the presence of an access made via the low-latency viewer (Step **S304**).

[0067] The switching unit **113** in the information processing device **100** then instructs the relaying unit **112** to switch to the parallel mode (Step **S305**).

[0068] Although the sending process of a stream from the camera **500** to the information processing device **100** (corresponding to Steps **S101** and **S102** in FIG. **2**) is omitted in FIG. **4**, this sending process may be continuously performed regardless of which mode the relaying unit **112** is in.

[0069] The relaying unit **112** relays the stream received from the transmitting unit **502** in the camera **500** after the instruction for switching to the parallel mode has been made, to both of the video server **210** and the low-latency server **220** (Step **S306** and Step **S307**).

[0070] The low-latency server **220** then transmits the stream to the receiving device **400** (Step **S308**). The stream relayed to the video server **210** can be distributed in the large-scale streaming process, in the same manner as in FIG. **3**, the process of which is not illustrated in FIG. **4**. In other words, in the parallel mode, the large-scale streaming process (FIG. **3**) and the low-latency transmit (Step **S308**, **S309**) are performed in parallel.

[0071] The receiving device **400** receives the stream from the low-latency server **220**, decodes the received stream, and displays the resultant stream, with a less delay time, using the low-latency viewer (Step **S309**).

[0072] An outline of a process for returning from the parallel mode to the normal mode will now be explained with reference to FIG. **5**. FIG. **5** is a flowchart giving an outline of the process for returning from the parallel mode to the normal mode in the first embodiment.

[0073] The user of the receiving device **400** then gives an instruction for ending the low-latency viewer, for example. The receiving device **400** then ends the low-latency viewer, in accordance with the ending instruction (Step **S401**).

[0074] In response to the ending of the low-latency viewer, the receiving device **400** disconnects the connection with the low-latency server **220** (Step **S402**).

[0075] When a plurality of low-latency viewers (the receiving devices **400**) are allowed to connect to the low-latency server **220**, the low-latency server **220** determines whether all of the low-latency viewers have been disconnected (Step **S403**). If not all of the low-latency viewers have been disconnected (No at Step **S403**), the process of returning from the parallel mode to the normal mode is aborted, and the parallel mode is maintained.

[0076] If all of the low-latency viewers have been disconnected (Yes at Step **S403**), the low-latency server **220** notifies the switching unit **113** in the information processing device **100** that there is no longer any access from the low-latency viewer (Step **S404**).

[0077] The switching unit **113** in the information processing device **100** instructs the relaying unit

112 to end the parallel mode, and to return to the normal mode (Step S405).

[0078] The relaying unit **112** relays the stream received from the transmitting unit **502** in the camera **500** after the instruction for switching to the normal mode has been made, only to the video server **210** (Step S406).

[0079] In the manner described above, in the first embodiment, the stream is relayed to both of the video server **210** for the large-scale streaming and the low-latency server **220** for the low-latency transmit, only when there is an access from the low-latency viewer. In this manner, it is possible to support both of the large-scale streaming and the low-latency transmit without interrupting the stream for the large-scale streaming, as well as without increasing the amount of data traffic from the camera **500**.

[0080] The relaying unit **112** and the low-latency server **220** do not need to be always kept running, and may be started only when the mode is switched to the parallel mode. For example, the switching unit **113** may start at least one of the relaying unit **112** and the low-latency server **220** when having received a sending request requesting to send a stream from the low-latency server **220** to the receiving device **400**.

[0081] Starting the relaying unit **112** means starting the execution of a computer program corresponding to the relaying unit **112**, or starting an operation of a circuit corresponding to the relaying unit **112**, on the information processing device **100**, for example. Starting the low-latency server **220** means starting the execution of a computer program corresponding to the low-latency server **220** on the cloud **200**, for example.

[0082] When the relaying unit **112** has not been started, e.g., during the normal mode, the transmitting unit **502** in the camera **500** may transmit the stream directly to the video server.

[0083] The switching unit **113** may also be provided to the camera **500**. For example, the camera **500** may include a switch (which may be hardware or software) for switching between the normal mode and the parallel mode. The switching unit **113** provided to the camera **500** instructs the information processing device **100** to switch between the normal mode and the parallel mode, in accordance with the switching of the switch. The switching unit **113** may switch the mode when a predetermined image (e.g., a QR code (registered trademark)) is captured with the camera **500**.

[0084] In such a case, the sending request requesting to send a stream from the low-latency server **220** to the receiving device **400** is sent from the camera **500** to the information processing device **100**. For example, when a sending request has been received from the camera **500**, the relaying unit **112** relays the stream ST1 to both of the video server **210** and the low-latency server **220**.

[0085] The switching unit **113** provided to the camera **500** may start at least one of the relaying unit **112** and the low-latency server **220** when the mode is switched to the parallel mode.

[0086] In the description so far, the viewer used for the large-scale streaming and the low-latency viewer have been distinguished, but it is also possible to use one viewer having both of these functions. In such a case, the receiving device **300** and the receiving device **400** may be integrated. For example, a user of the receiving device **300** (the receiving device **400**) accesses the web server **230** from a browser or the like, and receives a viewer including both of the functions of the large-scale streaming and the low-latency transmit from the web server **230**. Such a viewer is enabled to be designated with one of the large-scale streaming and the low-latency transmit using a radio button or the like, for example. When the large-scale streaming is designated, the viewer accesses the web server **230**, and receives and displays the stream via the large-scale streaming. When the low-latency transmit is designated, the viewer accesses the low-latency server **220**, and receives and displays the stream via a low-latency transmit.

[0087] The information processing system may be split into an information processing system SA including some of the devices, and an information processing system SB including the other devices. The information processing systems SA and SB may be built on respective cloud environments that are different from each other, for example. Exemplary configurations of the information processing system SA are listed below. [0088] (Exemplary configuration 1) An

information processing system SA including the information processing device **100** and the web server **230**. [0089] (Exemplary configuration 2) An information processing system SA including the information processing device **100**, the video server **210**, and the low-latency server **220**.

[0090] (Exemplary configuration 3) An information processing system SA including the information processing device **100** and the camera **500**.

[0091] In the manner described above, in the first embodiment, when there is an access to the low-latency server **220**, the stream is relayed to both of the video server **210** and the low-latency server **220**, and, when there is no access to the low-latency server **220**, the stream is relayed only to the video server **210**. This enables to continue the distribution of the stream while suppressing an increase in the data transmission quantity.

Second Embodiment

[0092] In the first embodiment, the same stream is transmitted in both of the normal mode and the parallel mode. In a second embodiment, during the parallel mode, when it is determined that it is not possible to achieve a low latency with the stream for the normal mode, encoding parameters are modified (adjusted) in such a manner that a low-latency sending is achieved.

[0093] FIG. **6** is a block diagram illustrating one example of a configuration of the information processing system according to the second embodiment. As illustrated in FIG. **6**, the information processing system according this embodiment includes an information processing device **100-2**, the cloud **200**, the receiving devices **300a**, **300b**, and **300c**, the receiving device **400**, and a camera **500-2**.

[0094] The information processing device **100-2** includes a processing unit **110-2**. The processing unit **110-2** includes the acquiring unit **111**, the relaying unit **112**, and a switching unit **113-2**. The camera **500-2** includes an encoding unit **501-2**, the transmitting unit **502**, a determining unit **503-2**, and an adjusting unit **504-2**.

[0095] The second embodiment is different from the first embodiment in further including a function of the switching unit **113-2**, a function of the encoding unit **501-2**, and in that the camera **500-2** is provided with the determining unit **503-2** and the adjusting unit **504-2**. The other configurations and functions are the same as those of the information processing device **100** according to the first embodiment illustrated in the block diagram of FIG. **1**, so the same reference numerals are given thereto, and explanations thereof will be omitted herein.

[0096] The switching unit **113-2** further includes a function for notifying the determining unit **503-2** that the mode has been switched to the parallel mode, upon switching the mode to the parallel mode.

[0097] The determining unit **503-2** determines whether it is possible to, with the stream captured during the normal mode, achieve a low-latency sending of the stream from the low-latency server **220**. For example, before switching to the parallel mode, the determining unit **503-2** determines whether the stream acquired (captured) by the camera **500** can be sent with a delay time equal to or less than a delay time **T2**.

[0098] If the determining unit **503-2** determines that the low-latency sending cannot be achieved, the adjusting unit **504-2** adjusts parameters on encoding performed by the encoding unit **501-2** (encoding parameters) in such a manner making the low-latency sending possible, and instructs the encoding unit **501-2** to use the adjusted encoding parameters.

[0099] The encoding unit **501-2** executes the encoding using the encoding parameters instructed by the adjusting unit **504-2**.

[0100] A method by which the determining unit **503-2** makes the determination and examples of the encoding parameters will be described later.

[0101] Various processes performed by the information processing device **100-2** according to the second embodiment will now be explained. Because the collecting process and the large-scale streaming process during the normal mode are the same as those in the first embodiment (FIG. **2** and FIG. **3**), explanations thereof will be omitted.

[0102] FIG. 7 is a flowchart giving the outline of a parallel mode switching process according to the second embodiment.

[0103] Because Steps S501 to S505 are the same as Steps S301 to S305 performed by the information processing device 100 according to first embodiment, explanations thereof will be omitted.

[0104] In this embodiment, the switching unit 113-2 notifies the determining unit 503-2 in the camera 500 that the mode has been switched to the parallel mode (Step S506). The information processing device 100-2 stores mapping information indicating mapping between the camera 500-2 and a stream, for example, in order to identify the camera that is currently transmitting the designated stream. The mapping information is information indicating the mapping between the address (e.g., IP address) of the camera 500-2 and a stream, for example. The information processing device 100-2 also stores therein information such as the address of the camera 500-2, for the purpose of sending an instruction to the camera 500-2.

[0105] The determining unit 503-2 in the camera 500-2 determines whether it is possible to achieve a low-latency sending, with the stream in the normal mode (Step S507). If it is not possible to achieve a low-latency sending (No at Step S507), the adjusting unit 504-2 adjusts the encoding parameters to those by which a low-latency sending can be achieved, and instructs the encoding unit 501-2 to perform encoding using the adjusted encoding parameters (Step S508).

[0106] After the encoding parameters are adjusted, or if it is determined that it is possible to achieve a low-latency sending (Yes at Step S507), the encoding unit 501-2 uses the instructed encoding parameters in the encoding of the stream captured from that time on (Step S509). The transmitting unit 502 then transmits the encoded stream to the information processing device 100-2 (Step S510).

[0107] Because Steps S511 to S514 are the same as Steps S306 to S309 performed by the information processing device 100 according to first embodiment, explanations thereof will be omitted.

[0108] An outline of the process for returning from the parallel mode to the normal mode in this embodiment will now be explained with reference to FIG. 8. FIG. 8 is a flowchart giving an outline of the process for returning from the parallel mode to the normal mode in the second embodiment.

[0109] Because Steps S601 to S605 are the same as Steps S401 to S405 performed by the information processing device 100 according to first embodiment, explanations thereof will be omitted.

[0110] When the parallel mode has been ended, the switching unit 113-2 notifies the determining unit 503-2 in the camera 500-2 that the parallel mode has been ended (Step S606).

[0111] The determining unit 503-2 notifies the adjusting unit 504-2 that the parallel mode has been ended. The adjusting unit 504-2 instructs the encoding unit 501-2 to change the encoding parameters to those of the normal mode (Step S607). If the encoding parameters have not been changed at Step S508 in FIG. 7, the adjusting unit 504-2 does not need to change the encoding parameters to those of the normal mode.

[0112] The encoding unit 501-2 uses the encoding parameters instructed by the adjusting unit 504-2 in the encoding of the stream captured from that time on (Step S608).

[0113] The transmitting unit 502 then transmits the encoded stream to the information processing device 100-2 (Step S609).

[0114] The relaying unit 112 in the information processing device 100-2 relays the stream received from the transmitting unit 502 only to the video server 210 (Step S610).

[0115] An example of a determining process performed by the determining unit 503-2 will now be explained. The determining unit 503-2 determines whether there will be any delay time in the transmit, that is, whether it is possible to achieve a low-latency sending, using a determining method such as those explained below. [0116] (Determining method 1) The time of the camera 500-2 is synchronized with the time of the information processing device 100-2. Transmission time

information is then embedded in a stream to be transmitted from the camera **500-2**, and the determining unit **503-2** is caused to calculate the delay time that is the difference between the sending time of the stream recorded in the camera **500-2** and the reception time of the stream recorded by the information processing device **100-2**. If the calculated delay time is within an acceptable range, the determining unit **503-2** determines that the low-latency sending is possible. [0117] (Determining method 2) A test program is installed in at least one of the camera **500-2** and the information processing device **100-2**. The test program then causes the camera **500-2** to send a piece of test data to the information processing device **100-2**, and estimates the maximum bit rate in the communication between the two. If a stream can be sent at a bit rate equal to or lower than the estimated maximum bit rate, the determining unit **503-2** determines that it is possible to achieve a low-latency sending.

[0118] An example of a process by which the adjusting unit **504-2** adjusts the encoding parameters will now be explained.

[0119] A stream includes an I frame containing information of the entire image frame, and a P frame and a B frame containing only the information of a difference with respect to a frame preceding or following the P frame or the B frame, for example. Because an I frame contains the entire information, the I frame has a large data size. Because a P frame and a B frame contain only the difference information, the P frame and the B frame have smaller data sizes.

[0120] As mentioned earlier, in HLS and DASH, the stream starts being replayed after waiting for the data corresponding to two segments to be buffered. Therefore, even if the time corresponding to two segments are used in the transmit of the stream, the latency in displaying the stream is not affected. By contrast, in the parallel mode, because there is no constraint of waiting for two segments, it may take time to transmit an I frame having a large data size over a low-speed network circuit, and displaying of the stream may become delayed. In other words, there are some cases in which a low-latency sending cannot be achieved.

[0121] Considering such a setting, if the determining unit **503-2** determines that a low-latency sending cannot be achieved, the adjusting unit **504-2** adjusts the encoding parameters using an adjusting method such as those described below. [0122] (Adjusting method 1) For example, if an I frame requires a transmit time longer than a threshold (if the delay time of the I frame exceeds the acceptable range), the adjusting unit **504-2** changes the encoding parameters so that I frames are divided and encoded in segments, using an intra-refresh mechanism, in order to reduce the time required in transmitting the I frame. [0123] (Adjusting method 2) For example, if an I frame requires a transmit time longer than a threshold, the adjusting unit **504-2** changes the encoding parameters by changing the balance in the amounts of stream encoded to the I frames and the P frames so as to make the data size of the I frames smaller. [0124] (Adjusting method 3) The adjusting unit **504-2** adjusts the encoding parameters by taking information of the network circuit to be used by the transmitting unit **502** into consideration. For example, when the network circuit is configured not to transmit data in excess of a specified bit rate, the adjusting unit **504-2** may adjust the maximum data size of the I frame, by adjusting the encoding parameters so that the specified bit rate is not exceeded.

[0125] The determining unit **503-2** and the adjusting unit **504-2** may be configured to repeat the determining process and the adjusting process until it is determined that a low-latency sending can be achieved.

[0126] There are times that the image quality deteriorates, as a result of adjusting the encoding parameters to enable a low-latency sending. To suppress such a deterioration in the image quality, the adjusting unit **504-2** may adjust the encoding parameters in a manner temporarily increasing the amount of encoding. For example, the adjusting unit **504-2** may set a maximum bit rate, and change the encoding parameters to those temporarily increasing the amount of encoding to maintain the image quality only while the parallel mode is selected, provided that communication at a bit rate equal to or lower than the maximum bit rate is possible.

First Modification

[0127] At least one of the determining unit **503-2** and the adjusting unit **504-2** may be provided to the information processing device **100-2**. FIG. **9** is a block diagram illustrating one example of a configuration of an information processing system according to a first modification.

[0128] As illustrated in FIG. **9**, the information processing system according to the first modification includes an information processing device **100-2b**, the cloud **200**, the receiving devices **300a**, **300b**, and **300c**, the receiving device **400**, and a camera **500-2b**.

[0129] In this modification, the camera **500-2b** includes neither the determining unit **503-2** nor the adjusting unit **504-2**, and the information processing device **100-2b** includes the determining unit **503-2** and the adjusting unit **504-2**. Because the other configurations are the same as those in the second embodiment, the same reference numerals are assigned thereto, and explanations thereof will be omitted herein.

[0130] As described above, in the second embodiment, when it is not possible to achieve the low-latency sending with a stream in the normal mode, encoding parameters can be changed only while there is an access from the low-latency viewer. In this manner, the large-scale streaming and the low-latency transmit can both be supported without increasing the amount of data traffic from the camera.

[0131] As described above, according to the first embodiment and the second embodiment, it is possible to continue streaming while suppressing an increase in the data transmission quantity.

[0132] A hardware configuration of the device described in the first or second embodiment (the information processing device, the receiving device, the video server, the low-latency server, the web server) will now be explained with reference to FIG. **10**. FIG. **10** is a schematic for explaining an example of the hardware configuration of the device according to the first or second embodiment.

[0133] The device according to the first or second embodiment includes a controller such as a central processing unit (CPU) **51**, a storage device such as a read-only memory (ROM) **52** and a random access memory (RAM) **53**, a communication interface (I/F) **54** connecting to a network and establishing communication, and a bus **61** connecting each of these units to one another.

[0134] A computer program executed by the device according to the first or second embodiment is provided in a manner incorporated in the ROM **52** or the like in advance.

[0135] The computer program executed by the device according to the first or second embodiment may also be provided as a computer program product that is recorded in a computer-readable recording medium such as a flexible disk (FD), a compact disc recordable (CD-R), or a digital versatile disc (DVD), as a file in an installable or executable format.

[0136] The computer program executed by the device according to the first or second embodiment may also be provided by storing the computer program in a computer connected to a network such as the Internet, and by making available for download over the network. The computer program executed by the device according to the first or second embodiment may also be provided or distributed over a network such as the Internet.

[0137] The computer program executed by the device according to the first or second embodiment can cause a computer to function as each of the units included in the device described above. The computer can cause the CPU **51** to read the computer program from a computer-readable storage medium onto the main memory, and to execute the computer program.

[0138] Configuration Examples of the embodiments are described below.

Configuration Example 1

[0139] An information processing device including a processing unit configured to: [0140] acquire a first stream transmitted from a transmitting device; and [0141] relay the acquired first stream to both a first server device and a second server device when a sending request requesting to send a stream from the second server device to a receiving device is received, and relay the acquired first stream to the first server device when the sending request is not received, the first server device

being configured to store an input stream into a storage device, the second server device being configured to send a stream with a delay time not greater than a second delay time that is less than a first delay time with which a stream stored in the storage device is distributed.

Configuration Example 2

[0142] The information processing device according to Configuration Example 1, wherein the processing unit includes: [0143] a relaying unit configured to relay the first stream; and [0144] a switching unit configured to start at least one of the relaying unit and the second server device when the sending request is received.

Configuration Example 3

[0145] The information processing device according to Configuration Example 1 or 2, wherein the processing unit is configured to relay, when the sending request is not received, the acquired first stream to the first server device but not to the second server device.

Configuration Example 4

[0146] The information processing device according to any one of Configuration Examples 1 to 3, wherein the processing unit is configured to: [0147] determine, when the sending request is received, whether it is possible to send the acquired stream with a delay time not greater than the second delay time; and [0148] adjust a parameter on encoding performed by an encoding unit configured to encode a stream, when it is determined that it is not possible to send the acquired stream with a delay time not greater than the second delay time.

Configuration Example 5

[0149] The information processing device according to Configuration Example 4, wherein the encoding unit is configured to perform encoding using an adjusted parameter.

Configuration Example 6

[0150] The information processing device according to any one of Configuration Examples 1 to 5, wherein [0151] the receiving device is a device configured to receive and display a stream sent from the second server device, and [0152] the processing unit is configured to relay, when the sending request is received from the receiving device, the acquired first stream to both the first server device and the second server device.

Configuration Example 7

[0153] The information processing device according to any one of Configuration Examples 1 to 6, wherein the processing unit is configured to relay, when the sending request is received from the transmitting device, the acquired first stream to both the first server device and the second server device.

Configuration Example 8

[0154] An information processing system including: [0155] the information processing device according to any one of Configuration Examples 1 to 7; and [0156] a third server device configured to acquire the first stream from the storage device in the first server device, and distribute the acquired first stream.

Configuration Example 9

[0157] An information processing system including: [0158] the information processing device according to any one of Configuration Examples 1 to 7; [0159] the first server device; and the second server device.

Configuration Example 10

[0160] An information processing system including: [0161] the information processing device according to Configuration Example 1; and [0162] the transmitting device.

Configuration Example 11

[0163] The information processing system according to Configuration Example 10, wherein the transmitting device includes: [0164] a determining unit configured to determine, when the sending request is received, whether it is possible to send the acquired stream with a delay time not greater than the second delay time; and [0165] an adjusting unit configured to adjust a parameter on

encoding performed by an encoding unit configured to encode a stream, when it is determined that it is not possible to send the acquired stream with a delay time not greater than the second delay time.

Configuration Example 12

[0166] The information processing system according to Configuration Example 11, wherein [0167] the encoding unit is configured to perform encoding using an adjusted parameter, and [0168] the transmitting device further includes a transmitting unit configured to transmit a stream encoded with an adjusted parameter to the information processing device.

Configuration Example 13

[0169] An information processing method executed by an information processing device, the information processing method including: [0170] acquiring a first stream transmitted from a transmitting device; and [0171] relaying the acquired first stream to both a first server device and a second server device when a sending request requesting to send a stream from the second server device to a receiving device is received, and relaying the acquired first stream to the first server device when the sending request is not received, the first server device being configured to store an input stream into a storage device, the second server device being configured to send a stream with a delay time not greater than a second delay time that is less than a first delay time with which a stream stored in the storage device is distributed.

Configuration Example 14

[0172] A computer program causing a computer to execute: [0173] acquiring a first stream transmitted from a transmitting device; and [0174] relaying the acquired first stream to both a first server device and a second server device when a sending request requesting to send a stream from the second server device to a receiving device is received, and relaying the acquired first stream to the first server device when the sending request is not received, the first server device being configured to store an input stream into a storage device, the second server device being configured to send a stream with a delay time not greater than a second delay time that is less than a first delay time with which a stream stored in the storage device is distributed.

[0175] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. An information processing device comprising one or more hardware processors configured to: acquire a first stream transmitted from a transmitting device; and relay the acquired first stream to both a first server device and a second server device when a sending request requesting to send a stream from the second server device to a receiving device is received, and relay the acquired first stream to the first server device when the sending request is not received, the first server device being configured to store an input stream into a storage device, the second server device being configured to send a stream with a delay time not greater than a second delay time that is less than a first delay time with which a stream stored in the storage device is distributed.
2. The device according to claim 1, wherein the one or more hardware processors include: one or more relaying hardware processors configured to relay the first stream; and one or more switching hardware processors configured to start at least either of the one or more relaying hardware processors and the second server device when the sending request is received.
3. The device according to claim 1, wherein the one or more hardware processors are configured to

relay, when the sending request is not received, the acquired first stream to the first server device but not to the second server device.

4. The information processing device according to claim 1, wherein the one or more hardware processors are configured to: determine, when the sending request is received, whether it is possible to send the acquired stream with a delay time not greater than the second delay time; and adjust a parameter on encoding performed by one or more encoding hardware processors configured to encode a stream, when it is determined that it is not possible to send the acquired stream with a delay time not greater than the second delay time.

5. The device according to claim 4, wherein the one or more encoding hardware processors is configured to perform encoding using an adjusted parameter.

6. The device according to claim 1, wherein the receiving device is a device configured to receive and display a stream sent from the second server device, and the one or more hardware processors are configured to relay, when the sending request is received from the receiving device, the acquired first stream to both the first server device and the second server device.

7. The device according to claim 1, wherein the one or more hardware processors are configured to relay, when the sending request is received from the transmitting device, the acquired first stream to both the first server device and the second server device.

8. An information processing system comprising: the information processing device according to claim 1; and a third server device configured to acquire the first stream from the storage device in the first server device, and distribute the acquired first stream.

9. An information processing system comprising: the information processing device according to claim 1; the first server device; and the second server device.

10. An information processing system comprising: the information processing device according to claim 1; and the transmitting device.

11. The system according to claim 10, wherein the transmitting device includes one or more hardware processors configured to: determine, when the sending request is received, whether it is possible to send the acquired stream with a delay time not greater than the second delay time; and adjust a parameter on encoding performed by one or more encoding hardware processors configured to encode a stream, when it is determined that it is not possible to send the acquired stream with a delay time not greater than the second delay time.

12. The system according to claim 11, wherein the one or more encoding hardware processors is configured to perform encoding using an adjusted parameter, and the transmitting device further includes one or more transmitting hardware processors configured to transmit a stream encoded with an adjusted parameter to the information processing device.

13. An information processing method executed by an information processing device, the information processing method comprising: acquiring a first stream transmitted from a transmitting device; and relaying the acquired first stream to both a first server device and a second server device when a sending request requesting to send a stream from the second server device to a receiving device is received, and relaying the acquired first stream to the first server device when the sending request is not received, the first server device being configured to store an input stream into a storage device, the second server device being configured to send a stream with a delay time not greater than a second delay time that is less than a first delay time with which a stream stored in the storage device is distributed.

14. A computer program product comprising a non-transitory computer-readable medium including programmed instructions, the instructions causing a computer to execute: acquiring a first stream transmitted from a transmitting device; and relaying the acquired first stream to both a first server device and a second server device when a sending request requesting to send a stream from the second server device to a receiving device is received, and relaying the acquired first stream to the first server device when the sending request is not received, the first server device being configured to store an input stream into a storage device, the second server device being configured to send a

stream with a delay time not greater than a second delay time that is less than a first delay time with which a stream stored in the storage device is distributed.
